

Quantum Measurement Stack (QMS)

A Three-Twin Architecture for Measurement Observability and Adaptive Inference

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QMS Platform v0.4.0

Software: <https://github.com/srikarr20/qms-platform>

License: Apache-2.0

Abstract

The Quantum Measurement Stack (QMS) is an open-source research and software framework for studying what can be inferred about a physical or quantum system from what its detector and measurement architecture make observable.

QMS has evolved from earlier work on detector-plane imaging and measurement observability into a provider-independent three-twin architecture comprising a Quantum Twin, Detector Twin, and Measurement Twin.

The Quantum Twin implements finite two-mode Gaussian dynamics, observability analysis, state reconstruction, Kalman estimation, physicality checks, parameter identification, model-family comparison, adaptive estimation, and measurement recommendation. The Detector Twin provides provider-independent residual analysis, stage-wise divergence assessment, inverse calibration, mechanism ranking, prediction, and detector-state comparison. The Measurement Twin operates directly on evolving observable measurements and provides causal short-horizon prediction, innovation analysis, frozen calibration, future-only alerting, and connected alert-episode consolidation.

Current validation includes controlled computational experiments, a Pyxel-based charge-transfer inefficiency parameter-recovery test, and causal-replay analysis of archival single-photon experimental measurements from the University of Glasgow.

The present evidence demonstrates internal computational consistency, finite-dimensional observability and estimation, detector-model parameter recovery under matched forward-model assumptions, and causal measurement-state tracking on real experimental data. It does not establish quantum-field reconstruction, universal quantum tomography, independently validated real-detector calibration, physical fault attribution, or prospective experimental validation.

The next validation stage is blind evaluation using independently configured detector simulations and held-out experimental measurements.

1. Introduction

Measurement systems do not provide direct access to an underlying physical state. They produce representations shaped by detector response, acquisition architecture, noise, sampling, processing, and model assumptions.

What can be inferred about an underlying physical or quantum system from what its measurement architecture actually makes observable?

The Quantum Measurement Stack was developed to investigate this question computationally.

The research lineage leading to QMS is:

Imprint Hypothesis -> Detector-Plane Imaging -> Measurement Observability -> QMS /
QMVA -> AURORA -> Adaptive Measurement Twin

The current QMS implementation consolidates these developments into a reusable architecture with three complementary twin layers.

2. Three-Twin Architecture

The current QMS architecture is organized as:

```
QMS
|
|-- Quantum Twin
|-- Detector Twin
`-- Measurement Twin
```

These layers address different inference questions while sharing a provider-independent measurement and inference philosophy.

2.1 Quantum Twin

What hidden finite-dimensional state could have produced the observed measurements, and is that state observable under the current measurement architecture?

The current implementation uses a finite two-mode Gaussian state representation:

```
x = [x1, p1, x2, p2]
```

Current capabilities include:

- linear Gaussian dynamics
- observability-matrix construction
- rank and null-space analysis
- conditioning analysis
- state reconstruction
- Kalman prediction and update
- Gaussian physicality checks
- residual and divergence analysis
- parameter identification
- model-family comparison
- adaptive parameter estimation
- measurement-configuration recommendation

The implementation is computational and finite-dimensional. It does not constitute general quantum-state tomography, quantum-field reconstruction, particle-trajectory inference, or experimentally validated quantum dynamics.

3. Detector Twin

What detector or measurement-chain behavior could explain the observed detector output?

The provider-independent detector layer includes:

- residual analysis
- stage-wise divergence detection
- mechanism ranking
- monotonic inverse calibration

- detector prediction
- innovation analysis
- detector-state assessment

Provider-specific detector physics are deliberately kept outside the QMS core. This separation allows QMS to interact with detector simulators or real detector systems without embedding a particular provider's implementation into the core architecture.

4. Measurement Twin

Is the observable measurement state evolving differently from what the current predictive model expects?

This layer does not require a fully specified detector-physics model. Current capabilities include:

- measurement-state construction
- cumulative-to-incremental reconstruction
- normalized measurement distributions
- cosine-similarity analysis
- Jensen-Shannon divergence
- constant-velocity state prediction
- innovation calculation
- frozen robust calibration
- future-only alerting
- connected alert-episode consolidation
- causal replay

This layer is intended to characterize unexpected measurement evolution without prematurely assigning a physical mechanism.

5. Pyxel Detector-Twin Integration

QMS includes a separate integration with the ESA-developed Pyxel detector-simulation framework. Pyxel provides the detector forward-model environment, while QMS provides observability, residual-analysis, inference, and adaptive-twin logic around the detector outputs.

A controlled charge-transfer inefficiency experiment was performed using Pyxel's charge-transfer model.

Quantity	Result
Known trap density	5.0e9 cm ⁻³
Recovered trap density	~4.99999948e9 cm ⁻³
Relative error	~1.05e-7
Pixel-stage residual RMSE	~1.53e-7

This result demonstrates numerical and inferential consistency within the tested Pyxel model family. It is a same-forward-model computational recovery test. It should not be interpreted as

independent detector validation, real-hardware calibration, or evidence of general detector-parameter inference.

5.1 Pyxel Citation

Arko, M., Prod'homme, T., Lemmel, F., Serra, B., George, E. M., Kelman, B., Pichon, T., Biancalani, E., & Gilbert, J. (2022). Pyxel 1.0: an open source Python framework for detector and end-to-end instrument simulation. *Journal of Astronomical Telescopes, Instruments, and Systems*, 8(4), 048002. DOI: <https://doi.org/10.1117/1.JATIS.8.4.048002>

Project: <https://esa.gitlab.io/pyxel/>

Source: <https://gitlab.com/esa/pyxel>

6. Glasgow Experimental Measurement-State Twin

QMS also includes a separate integration for archival single-photon experimental measurements from the University of Glasgow.

The current causal-replay analysis uses the archive:

Heralded Diffraction SM.zip

The workflow reconstructs incremental measurements from cumulative detector frames and then applies frozen-reference causal prediction and innovation analysis.

Configuration / outcome	Value
Reference end	399
Calibration end	699
Calibration samples	199
Median innovation	~0.00448429
MAD	~0.00229512
Frozen threshold	~0.02490077
Future samples	3369
Raw future flags	64
Connected alert episodes	6

Connected episode peak windows:

951, 1598, 2316, 2761, 3458, 3875

These are algorithmic connected clusters of temporally dependent alerts. They are not interpreted as statistically independent physical events, detector failures, or identified microscopic mechanisms.

The appropriate description of the present implementation is: causal-replay experimental measurement-state twin.

6.1 Glasgow Dataset Citation

Aspden, R. S., Padgett, M., & Spalding, G. (2016). Video recording true single-photon double-slit interference [Data collection]. University of Glasgow. DOI: <https://doi.org/10.5525/gla.researchdata.281>

Associated publication: Aspden, R. S., Padgett, M. J., & Spalding, G. C. (2016). Video recording true single-photon double-slit interference. *American Journal of Physics*, 84(9), 671-677. DOI: <https://doi.org/10.1119/1.4955173>

7. Software Architecture and Reproducibility

The current architecture separates reusable QMS logic from provider-specific integrations:

```
qms_core/
|-- quantum_twin/
|-- detector_twin/
`-- measurement_twin/

integrations/
|-- pyxel/
`-- glasgow/
```

Focused external-review packages are included under:

```
validation_packages/
|-- pyxel/
`-- glasgow/
```

The current test suite includes unit tests for the three QMS twin layers, the Pyxel CTI integration regression, and the Glasgow causal-replay integration regression.

Release-check test	Result
Main test suite	10 passed, 1 skipped
Pyxel external regression	1 passed
Glasgow external regression	1 passed

The externally dependent tests require access to their corresponding external data or simulation environments.

8. Scientific Boundaries

QMS v0.4.0 does not establish:

- quantum-field reconstruction
- general quantum-state tomography
- experimentally validated quantum dynamics
- universal detector-parameter inference
- independently validated real-detector calibration
- detector-health diagnosis
- physical fault attribution

- statistically independent physical-event discovery
- universal cross-instrument transfer
- prospective experimental validation

These boundaries are explicit because the present QMS objective is to build and validate the measurement and inference architecture before making stronger physical claims.

9. External Validation Roadmap

The strongest next experiments are:

1. independently configured Pyxel simulations whose parameters are hidden from QMS
2. model-family mismatch experiments
3. multiple competing detector mechanisms
4. held-out experimental detector sequences
5. frozen analysis parameters before test-data inspection
6. real detector calibration metadata where available
7. prospective streaming measurement evaluation

These experiments are intended to test where the QMS architecture fails as well as where it succeeds.

10. Conclusion

QMS v0.4.0 represents the transition from a collection of measurement-observability experiments into a provider-independent three-twin architecture.

The Quantum Twin addresses finite-dimensional hidden-state observability and estimation. The Detector Twin addresses model-based detector inference and residual behavior. The Measurement Twin addresses causal evolution of observable experimental measurement states.

Current evidence supports computational observability and estimation, same-model detector-parameter recovery, and causal-replay analysis on real experimental measurements.

The next major milestone is blind external validation.

Software Availability

Quantum Measurement Stack: <https://github.com/srikarr20/qms-platform>

Version: QMS Platform v0.4.0

License: Apache-2.0

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